Ethan Kolano

kolanoethan@gmail.com | (630) 398-1499 | ethan-kolano.github.io

Education

Bachelor's Degree in Mechanical Engineering Graduated May 2026
University of Illinois at Urbana-Champaign | GPA: 3.54

Technical Skills

CAD (Fusion 360), Python, Arduino, CNC Machining, Prototyping and Design, 3D Printing, Laser Cutting, Finite Element Analysis, Technical Drawings

Technical Projects

Rocky – Five-Legged Walking Robot May 2026 – Present
Personal Project

- Designing and building a 15-servo, five-legged walking robot in Fusion 360 with a 3D-printed structure, matching a film character's proportions as a fixed design constraint
- Exported the CAD assembly to a physics model with measured component masses and used PyBullet simulation to check stance stability and servo torque limits before hardware testing
- Assembled and tested the robot on Arduino-controlled hardware, resolving tipping and stability problems through iterative testing; it now stands, walks, and performs gestures

Ultrasonic Lens Cleaning System January 2026 – May 2026
Senior Design Project, sponsored by Carle Illinois College of Medicine | University of Illinois Urbana-Champaign

- Designed and fabricated a prototype integrating a PZT-8 annular piezo with a fused silica lens that successfully demonstrated ultrasonic water atomization
- Sized the lens thickness with a clamped-plate natural frequency calculation to match the 33 kHz piezo
- Selected bonding materials and structural components based on flexural strength, hardness, and elastic modulus to meet acoustic and mechanical targets

RC Car with Custom Transmission August 2025 – November 2025
Mechanical Design Course Project | University of Illinois Urbana-Champaign

- Designed and built a 3-gear sliding-mesh transmission with a servo-driven rack-and-pinion shifting mechanism, optimized for speed, strength, impact resistance, and manufacturability
- Performed structural and dynamic FEA on chassis and drivetrain to determine loads and optimize the design; fabricated parts by 3D printing and laser cutting

Walking Linkage Machine November 2024 – April 2025
ASME Engineering Open House Project | University of Illinois Urbana-Champaign

- Proposed and led a 9-member team to build a rideable, 8 ft Jansen-linkage walker on a $300 budget; won 1st place, Outstanding RSO Exhibit at Engineering Open House 2025
- Modeled the linkages in Fusion 360 to verify motion, and CNC-routed the plywood links

Experience

Undergraduate Researcher June 2025 – August 2025
Intelligent Machines & Materials Lab | Oregon State University

- Designed and prototyped a suction-based end-effector for a 6-axis robot arm for autonomous apple harvesting, integrating flex sensors for apple detection
- Developed an airtight housing with custom electrical feedthroughs and O-ring sealing to ensure reliable suction and signal integrity

Lighting Technician August 2023 – May 2026
Foellinger Auditorium | Urbana, IL

- Designed and operated lighting for performances, lectures, and events, working with directors and coordinators to meet each event's needs while a full-time engineering student